AI Multilingual Voice Studio

Introduction

AI Multilingual Voice Studio is a full-stack web-based application designed to provide real-time multilingual text translation, Hinglish conversion, and text-to-speech audio generation. The system integrates natural language processing and speech synthesis capabilities into a unified, user-friendly interface.

The application is intended for users who require quick translation along with audio output, making it useful for learning, accessibility, and communication purposes.

Objectives

The primary objectives of this project are:

- To develop a real-time multilingual translation system
- To provide audio output using text-to-speech technology
- To implement Hinglish conversion for improved usability
- To design an interactive and intuitive user interface
- To deploy a scalable cloud-based solution

System Overview

The application follows a client-server architecture:

- The frontend handles user interaction and UI rendering
- The backend processes requests, performs translation, and generates audio
- Communication occurs via REST API over HTTP

Features

The system provides the following functionalities:

- Multilingual text translation
- Hinglish text generation
- Text-to-speech audio output
- Audio file download capability
- Voice input using browser speech recognition
- Harmonium-based musical note generation from text
- Interactive keyboard-based music control

Technology Stack

Frontend:

- HTML5
- CSS3
- JavaScript
- Web Speech API

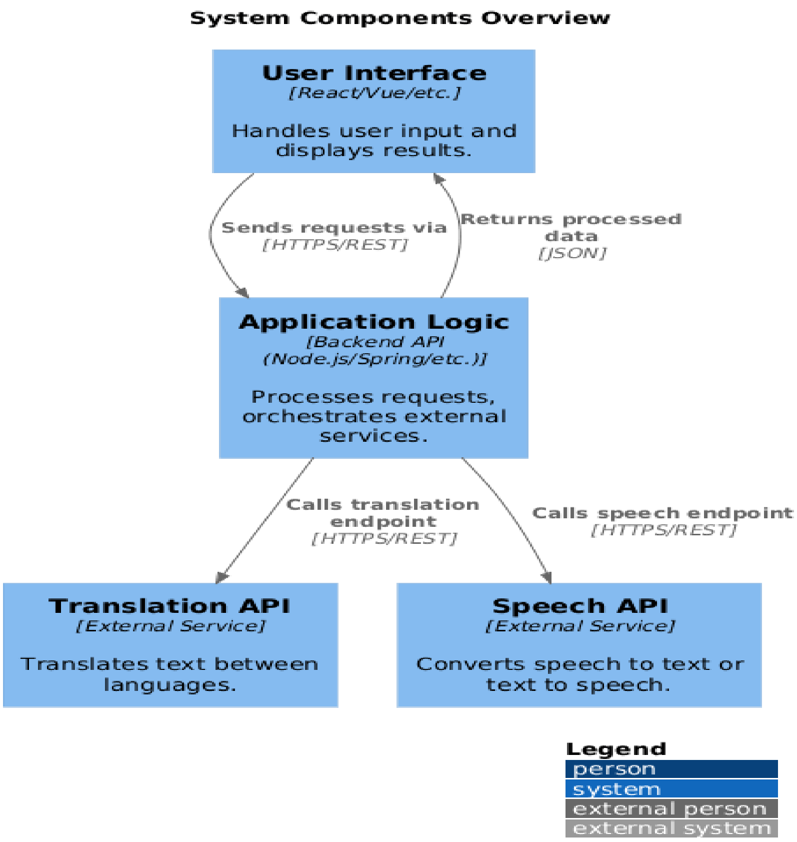
Backend:

- Python
- Flask
- gTTS (Google Text-to-Speech)
- deep-translator

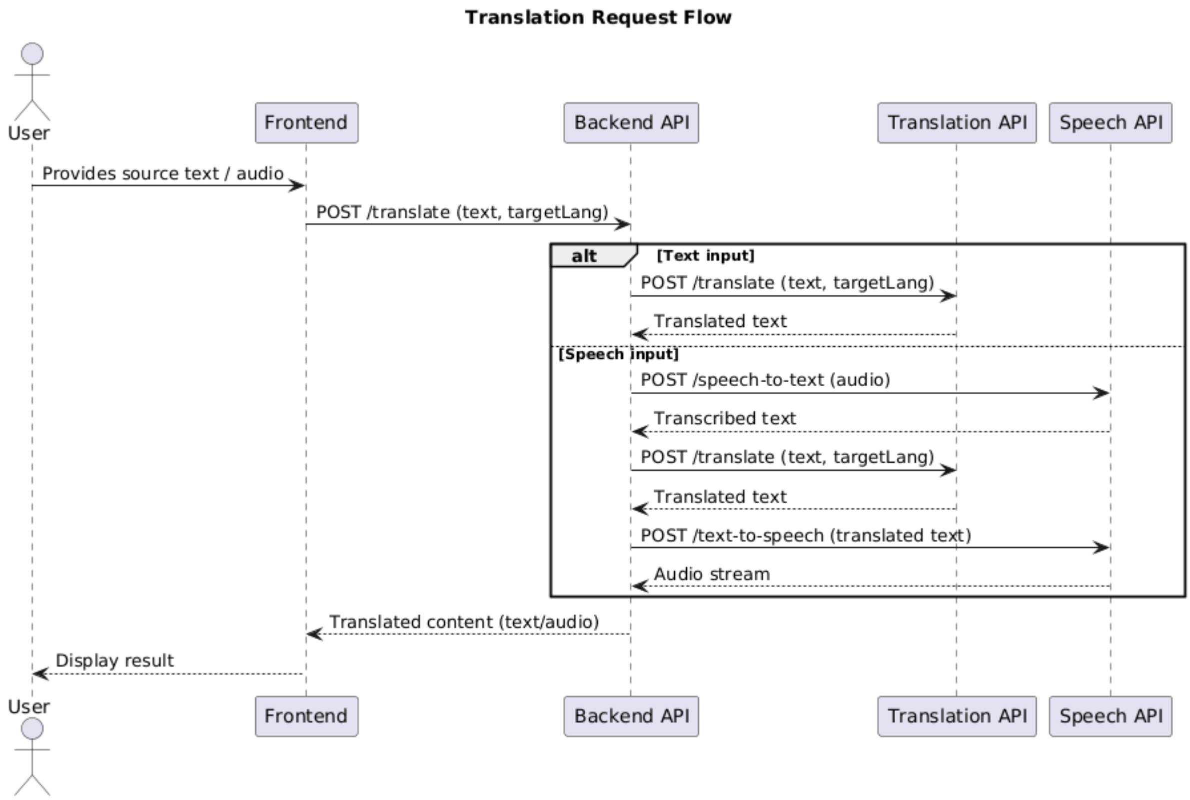
Deployment:

- Frontend hosted on Netlify
- Backend hosted on Render

System Architecture



Workflow



- User inputs text or voice
- Frontend sends request to backend API
- Backend processes translation and audio generation
- Response is returned to frontend
- UI updates with translated text and audio output

Project Structure

```
AI-Multilingual-Voice-Studio/
├── backend/
│   ├── app.py
│   ├── requirements.txt
│   └── audio_files/
├── frontend/
│   ├── index.html
│   └── sounds/
└── README.md
```

API Design

Endpoint: /process

Method: POST

Request Format:

```
{
  "text": "Input text",
  "lang": "target language code"
}
```

Response Format:

```
{
  "translated": "Translated text",
  "hinglish": "Hinglish version",
  "notes": ["sa", "re", "ga"],
  "audio_url": "URL to generated audio"
}
```

Functional Modules

9.1 Translation Module

Uses deep-translator to convert input text into the selected target language.

9.2 Hinglish Conversion Module

Transforms text into Hindi (interpreted as Hinglish representation).

9.3 Text-to-Speech Module

Generates audio using gTTS and stores it temporarily for playback and download.

9.4 Voice Input Module

Captures user speech using browser-based speech recognition.

9.5 Harmonium Module

Maps text characters to musical notes and plays them using preloaded audio files.

Deployment Strategy

Backend Deployment:

- Hosted on Render
- Uses Gunicorn as WSGI server
- Configured with requirements.txt

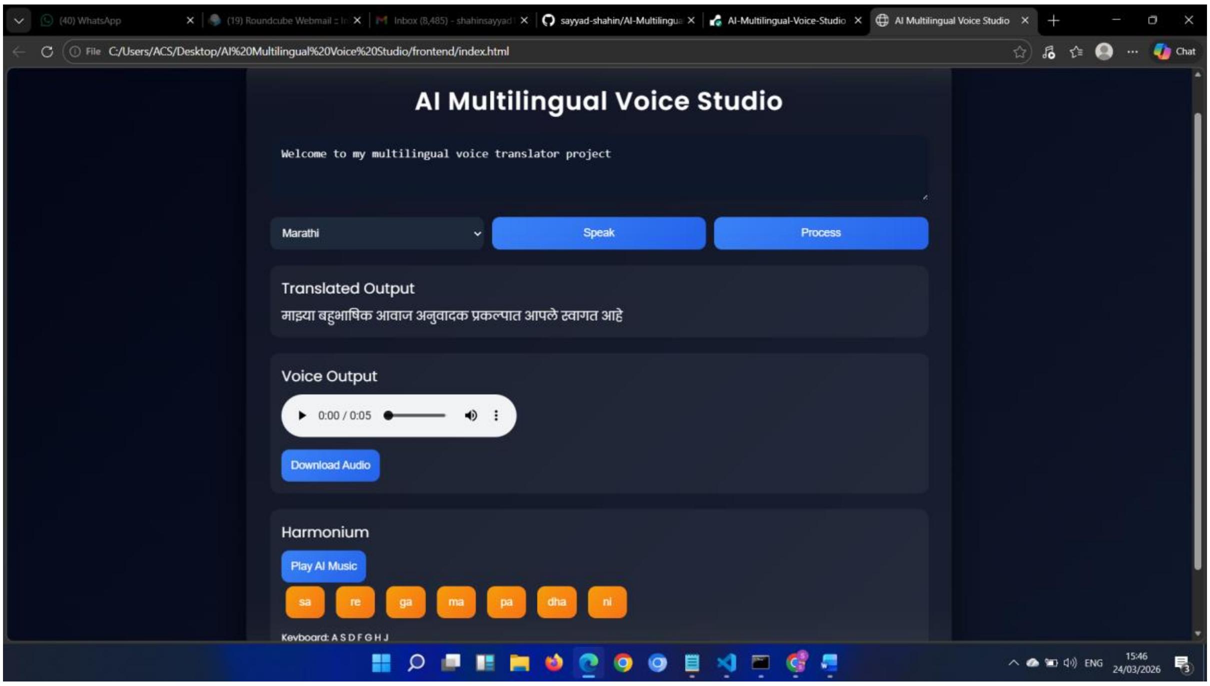
Frontend Deployment:

- Hosted on Netlify
- Deployed via drag-and-drop
- Connected to backend using public API endpoint

Performance Considerations

- Initial request latency due to free-tier hosting (cold start)
- Limited compute resources on free hosting plan
- Temporary storage of audio files
- Efficient handling of lightweight API requests

Implementation



Limitations

- Backend may experience delay after inactivity
- Audio files are not permanently stored
- Limited scalability on free hosting plan
- Hinglish conversion is simplified

Security Considerations

- CORS enabled for frontend-backend communication
- No authentication implemented (public access)
- Input validation is minimal
- No encryption for stored audio files

Future Enhancements

- Implement user authentication system
- Integrate advanced AI-based voice synthesis
- Add persistent storage for audio files
- Improve Hinglish conversion accuracy
- Enhance UI/UX with responsive design
- Add analytics and usage tracking
- Support real-time streaming audio

Conclusion

AI Multilingual Voice Studio demonstrates the integration of translation, speech synthesis, and interactive UI in a cloud-deployed full-stack application. The project highlights practical implementation of APIs, frontend-backend communication, and deployment strategies using modern platforms.